

ME 530: Continuum Mechanics

Course Code: ME 530

Course Name: Continuum Mechanics

L-T-P-C: 3-0-0-3

Prerequisites: Engineering Mechanics, Engineering Mathematics or equivalent

Students intended for: M.Tech./ M.S./ Ph.D./ B.Tech. 3rd and 4th year

Elective or Core: Elective

Mutual Exclusion: ME 530; CE 530 and ME 530 are same

Approval: 59th BoA

Course contents

- **Mathematical Essentials:** Scalars, Vectors and Cartesian Tensors; Indicical Notation; Tensor Algebra in Symbolic Notation: Summation Convention, Kronecker Delta, Permutation Symbol, $\epsilon - \delta$ Identity, Tensor/Vector Algebra; Matrices and Determinants; Transformations of Cartesian Tensors; Principal Values and Principal Directions; Tensor Fields, Tensor Calculus: Gradient and Divergence; Integral Theorems of Gauss and Stokes. [10 Lectures]
- **Kinematics of Deformation and Motion:** Particles, Configurations, Deformations and Motion; Material and Spatial Coordinates; Lagrangian and Eulerian Descriptions; Material Derivative; Displacement Field, Deformation Gradient, Finite Strain Tensors, Infinitesimal Deformation Theory; Compatibility Equations; Stretch Ratios; Rotation Tensor, Stretch Tensors; Velocity Gradient, Rate of Deformation, Vorticity, Steady and Irrotational Flow; Material Derivative of Line Elements, Areas, Volumes. [10 Lectures]
- **Stress Principle:** Body and Surface Forces, Cauchy Stress Principle, Stress Tensors: Cauchy and Piola-Kirchhoff Stress Tensors; Stress Transformation Laws; Principal Stresses, Principal Stress Directions, Mohr's Circles for Stress, Plane Stress; Deviator and Spherical Stress States; Octahedral Shear Stress. [8 Lectures]
- **Balance Laws and Equation:** Material Derivatives of Line, Surface and Volume Integrals; Conservation of Mass, Continuity Equation; Linear Momentum Principle, Equations of Motion in Lagrangian and Eulerian Descriptions; Moment of Momentum (Angular Momentum) Principle; Law of Conservation of Energy, The Energy Equation; Entropy and the Clausius-Duhem Equation; Governing Equations of a Continuum. [7 Lectures]
- **Constitutive Equations:** Constitutive Equations in Material Description; Linear Elasticity: Hooke's Law, Strain Energy, Elastic Symmetry, Hooke's Law for Anisotropic Media, Hooke's Law for Isotropic Media, Hyperelasticity; Nonlinear Elasticity: Strain Energy Theory for Nonlinear Elasticity, Specific Forms of the Strain Energy; Viscous Fluids: Viscous Stress Tensor, Stokesian, and Newtonian Fluids, Basic Equations of Viscous Flow. [7 Lectures]

Text Books:

1. G. Thomas Mase, Ronald E. Smelser, Jenn Stroud Rossmann, **Continuum Mechanics for Engineers**, CRC Press, 2020.



References:

1. Morton E. Gurtin, **An Introduction to Continuum Mechanics**, Academic Press, 1982.
2. Allan F. Bower, **Applied Mechanics of Solids**, CRC Press, 2009.
3. C. S. Jog, **Continuum Mechanics: Volume 1: Foundations and Applications of Mechanics**, Cambridge University Press, 2015.